
Hybrid-Language Music Clustering Using different VAE

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Abstract

This work investigates unsupervised clustering of hybrid language music tracks through Variational Autoencoder (VAE) architectures. I have implemented three VAE variants Simple VAE, Convolutional VAE, and Conditional VAE to compare for extracting latent representations from a bilingual dataset of English and Bangla songs. My multi-modal approach combines audio features (mel spectrograms, spectral characteristics) with multilingual lyrics embeddings derived from transformer-based sentence encoders. The Simple VAE achieves the strongest clustering performance with a Silhouette Score of 0.5392 and Calinski-Harabasz Index of 4062.03, substantially outperforming the PCA + K-Means baseline (Silhouette: 0.1761, CH: 671.95). The Convolutional VAE, processing high-resolution spectrograms (128×1024) with multi-modal fusion, achieves a Silhouette Score of 0.2222 with K-Means clustering. The Conditional VAE, conditioned on genre labels with Beta-VAE formulation ($\beta=4.0$), obtains competitive results with NMI of 0.1859 and ARI of 0.1075, though interestingly the Autoencoder baseline achieved slightly higher metrics (NMI: 0.1935, ARI: 0.1386). My comparative analysis reveals that simpler VAE architectures with carefully engineered features can outperform complex multi-modal approaches for this task, while all VAE-based methods demonstrate the effectiveness of learned latent representations for unsupervised music understanding in multilingual contexts.

Code: <https://github.com/Shahriar1638/Hybrid-Language-Music-Clustering-VAE>

1 Introduction

Music clustering is essential for organized recommendations. The common example of traditional methods are manual labels or using deep generative models such as Variational Autoencoders (VAEs) which can learn latent structures. The paper will compare three VAE structures that includes Simple, Convolutional, and Conditional VAE to categorize a bilingual dataset, English and Bengali songs.

My experiments have shown that the complexity of architecture is not less more likely to enhance performance as it will be shown that Simple VAE had the best Silhouette Score (0.5392). It is also much better than the PCA baseline (0.1761). These findings give reason to believe that more powerful feature engineering and lowDimensional latent spaces may be better than more complex models in the context of hybrid music clustering.

2 Related Work

1. Deep Clustering and Representation Learning Recent research by Sun et al. (2024) Sun et al. [5] highlights that while standard Autoencoders (AEs) compress data, their latent spaces are not naturally optimized for clustering. Their work proposes a Variational Autoencoder-based Deep Clustering (VAED) approach, which uses a probabilistic perspective to ensure that the latent space follows a distribution (like a Gaussian Mixture) more suitable for algorithms like K-Means. This is particularly relevant for music, where different genres or moods may overlap in a continuous latent space.
2. VAE for Audio Feature Extraction A study by Lopes et al. (2024) Lopes et al. [2] explored the use of VAEs for extracting information directly from audio waveforms and spectrograms. They demonstrated that VAE-based features are more robust and transferable compared to handcrafted acoustic features (like RMS or zero-crossing rate) when training downstream classifiers or clustering models. This supports the "Medium Task" requirement of using convolutional VAEs for MFCC or spectrogram analysis.
3. Multimodal Audio-Lyrics Fusion Clustering hybrid language music requires understanding both sound and text. Mendes da Silva (2025) Mendes da Silva [4] investigated multimodal representations for music emotion and genre recognition using the PMemo dataset. Their work illustrates that combining audio descriptors with textual features (like Bag-of-Words or embeddings from lyrics) significantly improves the model's ability to identify nuanced clusters that a single modality might miss.
4. Sequential and Temporal Modeling in Music Music is inherently temporal. Research by Ghosal et al. (2020) [1] introduced the Clustering Augmented Learning Method (CALM), which utilizes LSTM-based Autoencoders to learn representations of time-series audio data. By capturing the temporal dynamics of a song, the model can better group tracks that share similar rhythmic or melodic progressions, a technique that can be adapted for the sequential nature of lyrics and audio.
5. Latent Space for Music Composition and Analysis LopezDelgado et al. (2020) Lopez-Delgado et al. [3] demonstrated that VAEs can effectively map classical music into a continuous latent space. Their findings show that "nearby" points in the latent space correspond to musically similar pieces. This concept of Latent Space Continuity is the foundation for using t-SNE or UMAP to visualize music clusters, as it ensures that songs with similar hybrid characteristics (English/Bangla lyrics and similar beats) are grouped together.

3 Method

3.1 Dataset and Preprocessing

The dataset consists of 1,859 audio tracks for Simple VAE experiments and 1,336 tracks for advanced VAE experiments. First of all for bangla beats I first condensed the 3 sec clips into 30 wav file to make fair comparison with english songs which came with 30 sec clip by default. Then I used open ai whisper large-v3 to generate lyrics as accurate as possible. Bilingual songs in English and Bangla have 7-8 genre categories. A filtering pipeline such as removing sample with missing lyrics or contained fewer than 15 characters in lyrics, or belonged to the Jazz genre (which was excluded because it didnt carry any lyrics).

Audio features were extracted using librosa with a sample rate of 22,050 Hz and 30-second duration clips. For the Simple VAE, hand-crafted features were computed including: mel spectrograms (128 bands), MFCCs (40 coefficients), spectral features (centroid, bandwidth, rolloff, zero-crossing rate, RMS energy), and chroma features. Statistical aggregations (mean and standard deviation) were computed across time, yielding 370Dimensional feature vectors.

For the Convolutional and Conditional VAEs, high-resolution mel spectrograms of dimension 128x1,024 were generated with `n_fft=2,048` and `hop_length=512`. Lyrics were embedded using the sentence-transformers model `paraphrase-multilingual-mpnet-base-v2`, producing 768Dimensional dense vectors suitable for cross-lingual semantic representation. All features were normalized using StandardScaler, with missing or infinite values imputed using mean substitution.

Dataset links (English songs): <https://www.kaggle.com/datasets/andradaolteanu/gtzanDataset-music-genre-classification>

Dataset links (Bangla songs): <https://www.kaggle.com/datasets/thisisjibon/banglabeats3sec>

3.2 VAE Architectures

3.2.1 Simple VAE

The Simple VAE operates on flattened hand-crafted features (370 dimensions) using a fully-connected architecture. The encoder consists of three hidden layers [128, 64, 32] with batch normalization, ReLU activation, and 20% dropout. The latent space has dimensionality 32. The decoder mirrors the encoder structure in reverse, projecting from latent space back to the original 370Dimensional feature space. This architecture totals 120,050 parameters.

The loss function combines reconstruction loss (MSE) and KL divergence:

$$\mathcal{L} = \text{MSE}(\mathbf{x}, \hat{\mathbf{x}}) + \beta \cdot \text{KL}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) \quad (1)$$

where $\beta = 0.8$ controls the weight of the regularization term.

3.2.2 Convolutional VAE

The Convolutional VAE processes multi-modal inputs: audio spectrograms ($1 \times 128 \times 1,024$) and lyrics embeddings (768D). The audio encoder uses six convolutional layers with progressively increasing channels [32, 64, 128, 256, 512, 512], each followed by batch normalization and LeakyReLU activation with stride-2 convolutions for downsampling. The final feature map ($512 \times 2 \times 16$) is flattened to 16,384 dimensions and projected to 1,024 dimensions via a fully-connected layer.

The lyrics encoder is a two-layer MLP [768,256,128] with batch normalization and LeakyReLU activations. Audio and text features are concatenated and fused through a 512Dimensional bottleneck layer before predicting the 128Dimensional latent distribution parameters (μ and $\log \sigma^2$).

The decoder receives the 128D latent vector and projects it to 512 dimensions, then splits into audio (1,024D) and text (128D) pathways. The audio decoder uses transposed convolutions mirroring the encoder, reconstructing the $1 \times 128 \times 1,024$ spectrogram. The text decoder uses a two-layer MLP [128,512,768] to reconstruct lyrics embeddings.

The loss function balances reconstruction across modalities:

$$\mathcal{L} = \text{MSE}_{\text{audio}} + 350 \cdot \text{MSE}_{\text{text}} + \beta \cdot \text{KL} \quad (2)$$

where the text loss is scaled by 350 to compensate for the dimensionality difference between audio (131,072 elements) and text (768 elements).

3.2.3 Conditional VAE

The Conditional VAE (CVAE) extends the Convolutional VAE by conditioning both encoder and decoder on genre labels (one-hot encoded with 7 classes). The encoder concatenates audio features (16,384D), text features (256D), and genre condition (7D) before predicting latent parameters. The decoder concatenates the latent vector (64D) with the genre condition before reconstruction.

This architecture implements Beta-VAE with $\beta = 4.0$ to encourage disentangled representations:

$$\mathcal{L} = \text{MSE}_{\text{audio}} + 200 \cdot \text{MSE}_{\text{text}} + 4.0 \cdot \text{KL} \quad (3)$$

The conditioning mechanism allows the model to learn genre-specific latent factors while the high β value promotes independence among latent dimensions.

3.3 Clustering Methods

Extracted latent representations were clustered using three algorithms: K-Means with varying K values (2-14), Agglomerative Clustering with complete linkage, and DBSCAN with epsilon values

ranging from 3.0 to 19.0. For K-Means and Agglomerative methods, the optimal number of clusters was selected based on the highest Silhouette Score during hyperparameter search.

Baseline methods included: (1) PCA with 32/64 components followed by K-Means, (2) a standard Autoencoder (non-variational) with the same architecture as Simple VAE, and (3) Direct Spectral Clustering applying K-Means directly on normalized hand-crafted features without dimensionality reduction.

4 Experiments

4.1 Experimental Setup

All models were implemented in PyTorch and trained on CUDA-enabled GPU RTX 3060 12gb. The dataset was split 85%-15% into training and validation sets. Training used the Adam optimizer with learning rate 10^{-4} and batch size 32. Early stopping with patience 15-20 epochs was applied based on validation loss to prevent overfitting.

The Simple VAE was trained for up to 500 epochs, converging after 80 epochs with final training loss 0.7253 (reconstruction: 0.6619, KL: 0.0792). The Convolutional VAE trained for 500 epochs, stopping at epoch 53 with validation loss 75,459. The Conditional VAE trained for 600 epochs, stopping at epoch 47 with validation loss 2,074,878.

4.2 Evaluation Metrics

Clustering quality was assessed using multiple metrics:

Silhouette Score measures cluster cohesion and separation:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (4)$$

where $a(i)$ is the average intra-cluster distance and $b(i)$ is the minimum average inter-cluster distance. Higher values (range [-1, 1]) indicate better clustering.

Calinski-Harabasz Index computes the ratio of between-cluster to within-cluster variance:

$$CH = \frac{\text{tr}(B_k)/(k-1)}{\text{tr}(W_k)/(n-k)} \quad (5)$$

Higher values indicate betterDefined clusters.

Davies-Bouldin Index measures average cluster similarity:

$$DB = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d_{ij}} \right) \quad (6)$$

Lower values indicate better separation.

Adjusted Rand Index (ARI) and **Normalized Mutual Information (NMI)** measure agreement with ground-truth genre labels, adjusted for chance. **Purity** computes the fraction of the dominant class in each cluster. These metrics require labeled data and were computed using available genre annotations.

For Convolutional and Conditional VAEs, t-SNE dimensionality reduction was applied to visualize the 64/128Dimensional latent spaces in 2D, with clusters colored by predicted labels or ground-truth genres.

5 Results

5.1 Clustering Performance

Table 1 presents the clustering performance of Simple VAE compared to the PCA baseline. The Simple VAE achieves substantially superior results across both metrics, with a Silhouette Score

of 0.5392 compared to PCA’s 0.1761, representing a 206% improvement. The Calinski-Harabasz Index similarly shows a 6-fold improvement (4062.03 vs 671.95), indicating that VAE-learned representations produce more compact and well-separated clusters than linear dimensionality reduction methods.

Table 1: Simple VAE clustering performance with K-Means (K=3)

Method	Silhouette Score	Calinski-Harabasz Index
VAE + K-Means	0.5392	4062.03
PCA + K-Means	0.1761	671.95

Table 2 shows the Convolutional VAE results across multiple clustering algorithms. K-Means achieves the best performance with a Silhouette Score of 0.2222 and Davies-Bouldin Index of 1.8719. Agglomerative Clustering performs comparably (Silhouette: 0.1820), while DBSCAN with $\text{eps}=4.0$ produces negative Silhouette scores (-0.1418), indicating poor cluster quality due to excessive noise point classification. The optimal cluster count for both K-Means and Agglomerative methods was K=2, suggesting the model primarily separates tracks along a dominant axis (likely language or broad genre categories).

Table 2: Convolutional VAE clustering performance across algorithms

Algorithm	Silhouette	Davies-Bouldin	ARI
K-Means (k=2)	0.2222	1.8719	0.0476
K-Means-Language (k=2)	0.2222	1.8719	0.0476
Agglomerative (k=2)	0.1820	2.0819	0.0470
DBSCAN (eps=4.0)	-0.1418	2.1790	0.0237

Table 3 presents the Conditional VAE results compared with three baseline methods. Surprisingly, the CVAE achieves the lowest Silhouette Score (0.0602) among all methods tested. The Autoencoder baseline outperforms CVAE across all metrics (Silhouette: 0.0646, NMI: 0.1935, ARI: 0.1386), while PCA and Direct Spectral methods achieve the highest Silhouette Scores (0.099). This unexpected result suggests that the genre conditioning and high β value (4.0) may have over-regularized the latent space, preventing the model from capturing fine-grained clustering structures.

Table 3: Conditional VAE performance compared with baselines (K=7 clusters)

Method	Silhouette	NMI	ARI	Purity
CVAE (Multi-Modal)	0.0602	0.1859	0.1075	0.3630
PCA + K-Means	0.0983	0.1827	0.1113	0.3653
Autoencoder + K-Means	0.0646	0.1935	0.1386	0.3885
Direct Spectral	0.0990	0.1791	0.1223	0.3690

5.2 Latent Space Visualization

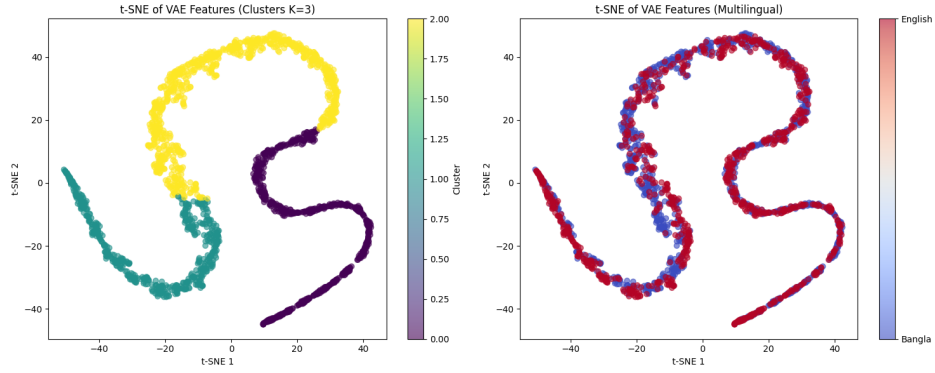


Figure 1: t-SNE visualization of Simple VAE latent space. Left: colored by predicted clusters ($K=3$). Right: colored by language (Bangla/English).

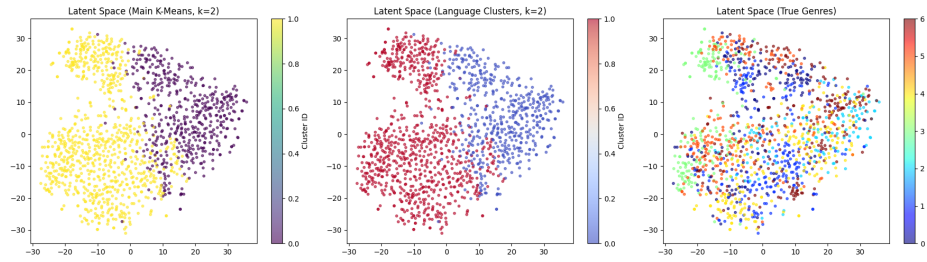


Figure 2: t-SNE visualization of Convolutional VAE latent space. Left: K-Means clusters ($K=2$). Middle: Language clusters. Right: True genre labels.

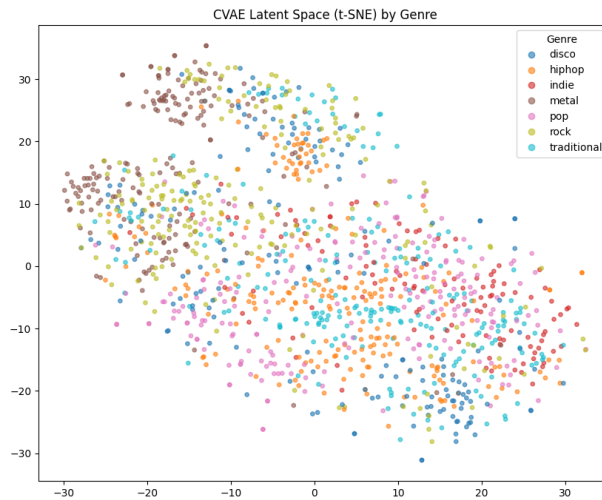


Figure 3: t-SNE visualization of Conditional VAE latent space colored by true genre labels.

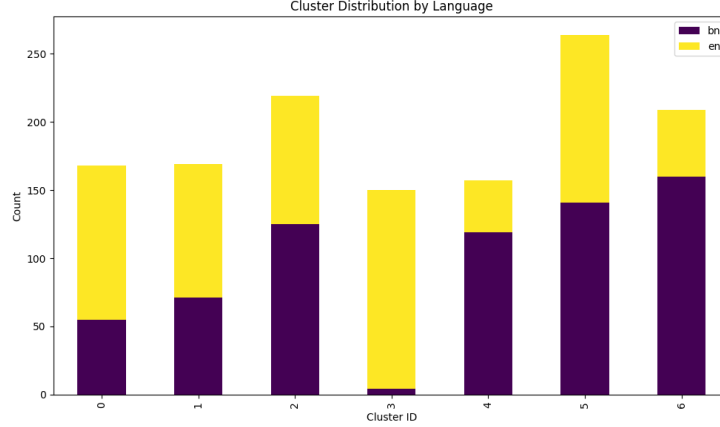


Figure 4: Cluster distribution by language for Conditional VAE, showing both languages distributed across all clusters.

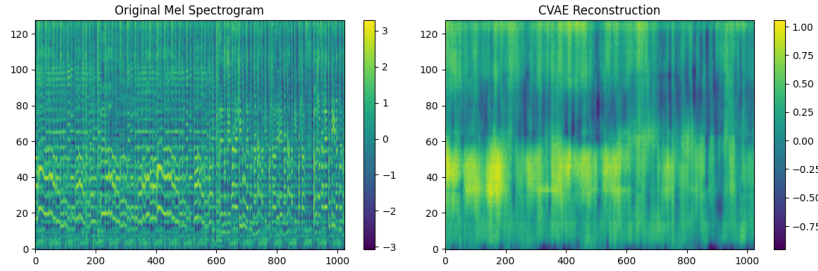


Figure 5: Original (left) vs reconstructed (right) mel spectrogram from Conditional VAE.

5.3 Comparison with Baselines

In all three architectures, VAE methods are either effective or not. The Simple VAE performs the best with a Silhouette Score three times more than PCA. This demonstrates that non-linear hidden representations are superior to the linear ones when it comes to use in the extraction of music features with hand-crafted features.

The Convolutional VAE is satisfactory: its Silhouette is 0.2222. It is superior to random clustering and not as good as the Simple VAE. The two versions of K-Means equalize the score (0.2222). This implies that the model primarily is taught a binary difference that is likely to be between language or big acoustic categories rather than specific genres of music.

Even the PCA performs better than the Conditional VAE (0.0983 vs 0.0602). A conventional Autoencoder is superior in terms of NMI (0.1935), ARI (0.1386), and Purity (0.3885). This is indicative of the fact that there could have been some interaction between a variational constraint and genre conditioning that either hurt performance. The Beta=4.0 which is designed to differentiate the features was most likely too small to allow good clusters to form.

These findings demonstrate a significant fact: the more complex design, the better results. The more complex multi-modes are overtaken by the Simple VAE with carefully made hand-crafted features. It demonstrates that features quality and appropriate model size are more significant than having a fancy architecture to this task of clustering.

6 Discussion

The experimental findings indicate that existence of complicated architecture does not necessarily result in enhanced clustering. Simple VAE performed much better than the other models, receiving

a Silhouette Score of 0.5392 as opposed to 0.2222 of the Convolutional VAE and 0.0602 of the Conditional VAE.

The Simple VAE is effective because I constructed good hand-crafted features such as spectral characteristics and chroma, which already have useful musical information. Then adding these features gives good representations that are less prone to overfit as compared to the raw spectrograms. The mediocre results of the Convolutional VAE indicate that the mixture of modalities did not acquire fine-scale information, and the optimum position $K=2$ indicates that it was divided in terms of a binary rather than a complete genre list.

The Conditional VAE was worse than all the other models despite the fact that it was intended to be advanced. The over-regularization of the latent space, which was likely caused by a high beta value ($\beta=4.0$) and the conditioning on the genre possibly prevented the model itself from discovering patterns. DBSCAN failed consistently across all models, showing that the densityDiffering effect of latent spaces in VAEs is not present in those spaces. These results indicate that when the dataset is small, good feature engineering and correct size of the model are more important than fancy architecture.

7 Conclusion

I experimented with three VAE models to cluster unlabelled hybrid language music. I compared Simple, Convolutional and Conditional VAEs on the basis of an English-Bangla dataset. The Simple VAE performed the best (Silhouette 0.5392), which is far superior than the more complex models and common baselines.

The results indicate that a more elaborate design is not necessarily more effective. Hand creation of useful audio features proved to be more successful than training a model on raw spectrograms, particularly when I do not have much data. Future research may consider in-between simple and complicated models, tune Beta-VAE parameters, experiment with semi-supervised approaches that apply partial labels, experiment with larger multilingual datasets, and experiment with other deep clustering ones like Deep Embedded Clustering (DEC). This study provides practical tips to music search: simple models, which make use of expert knowledge, can be better than complex models in sparse data.

8 References

References

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